

Metadata Management Standards V2.2

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APPROVAL SIGNATURES

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1. OBJECTIVE AND SCOPE

1.1 Purpose

This document establishes mandatory standards for metadata collection, documentation, and maintenance to ensure every data asset at Company A is discoverable, understood, and trusted. High-quality metadata enables:

- Data discovery and self-service analytics
- Impact analysis for system changes
- Regulatory compliance documentation
- Data quality management
- Effective data governance

1.2 Strategic Goal

"No Data Left Behind" - 100% of production data assets must be described with complete, accurate metadata meeting the standards defined in this document.

1.3 Scope

These standards apply to:

- All production databases and data stores
- All reporting and analytics assets
- All data integration processes (ETL/ELT)
- All APIs and data services
- All master and reference data
- All external data sources

1.4 Metadata Repository

Component	Tool	Purpose
Enterprise Data Catalog	Informatica EDC	Central metadata repository
Business Glossary	Informatica EDC	Business term definitions
Data Lineage	Informatica EDC	End-to-end data flow
Data Profiling	Informatica IDQ	Technical metadata statistics
API Catalog	Azure API Management	API documentation

1.5 NDMO Alignment

NDMO Control	Requirement	Implementation
DC-01	Metadata repository	Informatica EDC
DC-02	Business glossary	Section 3 standards
DC-03	Technical metadata	Section 4 standards
DC-04	Data lineage	Section 4.4 standards
DC-05	Metadata governance	Section 6 standards

2. METADATA TYPES

2.1 Business Metadata

Definition

Information that provides business context to technical data assets, enabling business users to understand and trust the data.

Components

Component	Description	Example
Business Terms	Standardized vocabulary	"Patient", "Encounter", "Diagnosis"
Definitions	Clear explanations	"Patient: An individual receiving healthcare servi
Policies	Applicable rules	"PDPL applies to patient data"
Business Rules	Data validation logic	"Patient age must be 0-150 years"
Data Owners	Accountable persons	"Dr. Sarah Al-Qahtani - CMO"

Audience

- Business analysts
- Data consumers
- Compliance officers
- Executive leadership

2.2 Technical Metadata

Definition

Information describing the physical characteristics and structure of data assets.

Components

Component	Description	Example
Schema	Database structure	"EPIC_PROD.PATIENTS"
Tables	Data containers	"TBL_PATIENT_DEMOGRAPHICS"
Columns	Field definitions	"PATIENT_ID NUMBER(10)"
Data Types	Storage formats	"VARCHAR2, DATE, NUMBER"
Indexes	Performance structures	"IDX_PATIENT_MRN"
Constraints	Integrity rules	"PK_PATIENT_ID PRIMARY KEY"
Relationships	Table connections	"PATIENT_ID FK to ENCOUNTERS"

Audience

- Database administrators
- Application developers
- Data engineers
- IT support teams

2.3 Operational Metadata

Definition

Information about data processing activities, performance metrics, and runtime statistics.

Components

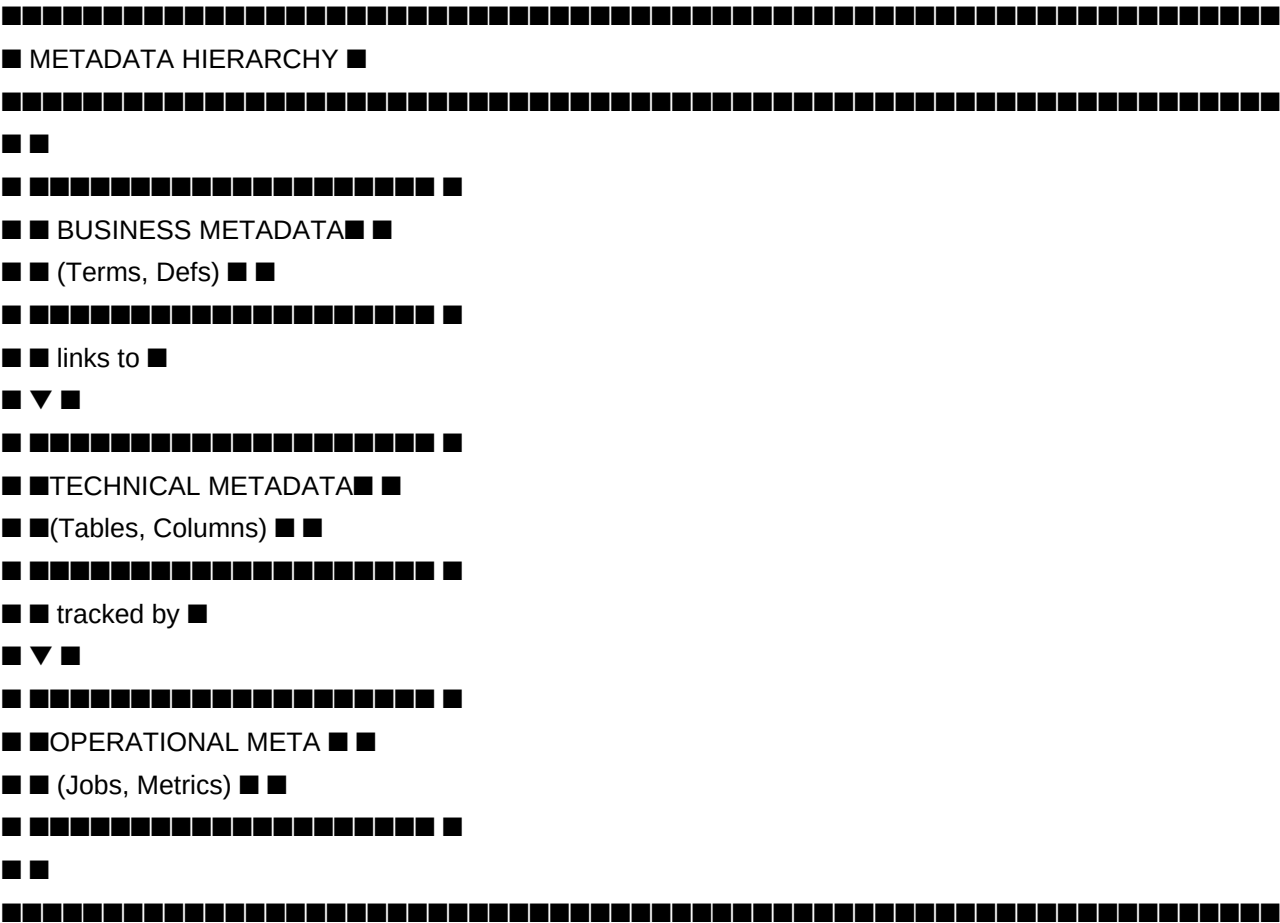
Component	Description	Example
Job Schedules	When processes run	"Daily 02:00 AM"
Run History	Execution records	"Last run: 2024-06-01 02:15:00"

Row Counts	Volume metrics	"Loaded: 15,234 rows"
Success/Fail Status	Outcome tracking	"Status: SUCCESS"
Duration	Performance metrics	"Duration: 45 minutes"
Error Logs	Issue documentation	"Warning: 12 duplicate keys"

Audience

- Operations teams
- Support analysts
- Performance engineers
- Data stewards

2.4 Metadata Relationships



3. BUSINESS METADATA STANDARDS

3.1 The "5 Ws" Rule

Every business term definition must answer the following questions:

Question	Description	Example (Patient)
What	What is this data element?	An individual receiving healthcare services
Where	Where does it come from?	Created at registration desk
When	When is it updated?	Updated at each encounter
Who	Who owns/manages it?	Registration Department
Why	Why do we collect it?	Required for clinical care delivery

3.2 Definition Quality Standards

3.2.1 Length Requirements

Definition Type	Minimum	Maximum
Business Term	20 characters	500 characters
Column Description	15 characters	250 characters
Table Description	50 characters	1000 characters

3.2.2 Clarity Requirements

Requirement	Bad Example	Good Example
No recursive definitions	"Customer ID is the ID of the Customer"	"A unique 10-digit number assigned to an individual"
No jargon without explanation	"MRN"	"Medical Record Number (MRN): Unique identifier for patient"
Action-oriented for processes	"ETL"	"Extract patient data from Epic, transform formats, load into data warehouse"

3.2.3 Acronym Standards

All acronyms must be spelled out on first use:

Acronym	Full Form
DOB	Date of Birth
MRN	Medical Record Number
SSN	Saudi Social Number
ICD	International Classification of Diseases

CPT	Current Procedural Terminology
HL7	Health Level Seven
FHIR	Fast Healthcare Interoperability Resources

3.3 Tagging Taxonomy

3.3.1 Confidentiality Tags

Tag	Color	Description
PUBLIC	Green	Openly shareable
INTERNAL	Blue	Company use only
CONFIDENTIAL	Orange	Restricted access
SECRET	Red	Highly restricted

3.3.2 Domain Tags

Tag	Department	Data Examples
HR	Human Resources	Employee records, payroll
FIN	Finance	Invoices, budgets, GL
MED	Medical/Clinical	Patient records, diagnoses
OPS	Operations	Facilities, logistics
IT	Information Technology	Systems, logs, configs
SEC	Security	Access logs, incidents

3.3.3 Regulatory Tags

Tag	Regulation	Implication
PDPL	Personal Data Protection Law	Privacy controls required
CBAHI	Healthcare accreditation	Medical record standards
ZATCA	Tax authority	Financial retention rules
SAMA	Financial regulation	Banking data controls
NCA	Cybersecurity authority	Security requirements

3.4 Business Term Governance

3.4.1 Term Lifecycle

Status	Description	Next Action
DRAFT	Initial creation	Submit for review
REVIEW	Under evaluation	Approve or reject
APPROVED	Standard term	Publish to glossary
DEPRECATED	No longer preferred	Migrate to replacement
ARCHIVED	Historical reference	No active use

3.4.2 Term Ownership

Role	Responsibility
Term Owner	Business accountability for accuracy
Term Steward	Day-to-day maintenance
Subject Matter Expert	Validation of definition
Data Architect	Technical alignment

4. TECHNICAL METADATA STANDARDS

4.1 Database Object Naming Conventions

4.1.1 Table Naming

Convention	Pattern	Example
Prefix by type	TBL_, VW_, MV_	TBL_PATIENTS
Uppercase	SNAKE_CASE	PATIENT_DEMOGRAPHICS
Singular nouns	Not plural	TBL_PATIENT (not TBL_PATIENTS)
No abbreviations	Full words	ENCOUNTER (not ENCNTN)

4.1.2 Column Naming

Convention	Pattern	Example
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Descriptive	Clear meaning	PATIENT_FIRST_NAME
Suffix for dates	_DATE, _DATETIME	ADMISSION_DATE
Suffix for IDs	_ID, _CODE	PATIENT_ID
Suffix for flags	_FLAG, _IND	ACTIVE_FLAG
Suffix for amounts	_AMT, _AMOUNT	INVOICE_AMOUNT

4.1.3 Index Naming

Convention	Pattern	Example
Primary Key	PK_tablename	PK_PATIENT
Foreign Key	FK_table_reftable	FK_ENCOUNTER_PATIENT
Unique Index	UK_table_columns	UK_PATIENT_MRN
Non-unique	IDX_table_columns	IDX_PATIENT_DOB

4.2 Custom Metadata Attributes

Company A extends the standard metadata model with the following custom attributes:

Attribute	Description	Values
Original_Legacy_Name	Name in legacy 2010 system	Free text
Cost_Center_Code	Budget owner	e.g., "CC-IT-001"
Data_Classification	Security level	PUBLIC/INTERNAL/CONFIDENTIAL/SECRET
PII_Indicator	Contains personal data	YES/NO
PHI_Indicator	Contains health data	YES/NO
Retention_Period	How long to keep	e.g., "10 years"
Source_System	Origin system	e.g., "Epic", "Oracle EBS"
Refresh_Frequency	Update schedule	REALTIME/DAILY/WEEKLY/MONTHLY

4.3 Data Profiling Standards

4.3.1 Mandatory Profiling Metrics

Metric	Description	Tier 1 Required	Tier 2 Required
Row Count	Total records	Yes	Yes

Null Count	Missing values	Yes	Yes
Null Percentage	% missing	Yes	Yes
Distinct Count	Unique values	Yes	Optional
Min Value	Minimum	Yes	Optional
Max Value	Maximum	Yes	Optional
Mean Value	Average (numeric)	Yes	No
Pattern Analysis	Format detection	Yes	No
Value Frequency	Top N values	Yes	No

4.3.2 Profiling Frequency

System Tier	Profiling Frequency	Tool
Tier 0	Weekly	Informatica IDQ
Tier 1	Weekly	Informatica IDQ
Tier 2	Monthly	Informatica IDQ
Tier 3	Quarterly	Informatica IDQ

4.4 Data Lineage Standards

4.4.1 Lineage Capture Requirements

Lineage Type	Scope	Tool
Source-to-Target	ETL jobs	Informatica EDC Scanner
Column-level	Field mapping	Informatica EDC
Transformation	Business logic	ETL documentation
Report-to-Source	BI assets	Tableau/Power BI metadata

4.4.2 Lineage Documentation

Each data flow must document:

Element	Description
Source System	Origin of data
Source Object	Table/file name
Transformation Logic	Business rules applied

Target System	Destination
Target Object	Table/file name
Refresh Method	Full/Incremental
Schedule	When it runs

5. OPERATIONAL METADATA STANDARDS

5.1 Job Metadata Requirements

Attribute	Description	Mandatory
Job Name	Unique identifier	Yes
Job Description	Purpose explanation	Yes
Schedule	Cron expression or description	Yes
Owner	Responsible team/person	Yes
Source Systems	Input data sources	Yes
Target Systems	Output destinations	Yes
Dependencies	Prerequisite jobs	Yes
SLA	Expected completion time	Yes
Alert Contacts	Who to notify on failure	Yes

5.2 Execution Logging Standards

5.2.1 Required Log Elements

Element	Description
Job ID	Unique execution identifier
Start Time	When job began
End Time	When job completed
Status	SUCCESS/FAILURE/WARNING
Records Read	Input count
Records Written	Output count
Records Rejected	Failed records

Error Messages	Failure details
Duration	Elapsed time

5.2.2 Log Retention

Log Type	Retention Period
Success logs	90 days
Failure logs	1 year
Performance metrics	2 years

6. METADATA COLLECTION AND MAINTENANCE

6.1 Automated Collection

Source Type	Collection Method	Tool	Frequency
Oracle Databases	JDBC Scanner	Informatica EDC	Daily
SQL Server	JDBC Scanner	Informatica EDC	Daily
PostgreSQL	JDBC Scanner	Informatica EDC	Daily
Informatica Jobs	Lineage Extraction	Informatica EDC	Daily
Tableau Workbooks	REST API	Informatica EDC	Daily
Power BI Reports	Graph API	Informatica EDC	Daily
API Specifications	OpenAPI Import	Azure APIM	On publish

6.2 Manual Enrichment

Metadata Type	Responsibility	Frequency
Business definitions	Data Steward	On creation/change
Data owner assignment	Data Governance	On creation
Classification tags	Data Steward	On creation
Regulatory tags	Compliance	On creation/audit

6.3 Maintenance Schedule

Activity	Frequency	Responsibility
Scanner execution	Daily	Automated
New asset review	Weekly	Data Steward
Definition accuracy check	Monthly	Data Steward
Orphan cleanup	Monthly	Data Architect
Completeness audit	Quarterly	Data Governance

7. METADATA QUALITY REQUIREMENTS

7.1 Quality Dimensions

Dimension	Definition	Target
Completeness	All required attributes populated	> 95%
Accuracy	Information is correct	> 98%
Consistency	No conflicting metadata	100%
Timeliness	Metadata is current	< 7 days lag

7.2 Quality Thresholds by Asset Type

Asset Type	Completeness	Accuracy
Tier 1 Tables	100%	100%
Tier 2 Tables	95%	98%
Reports	90%	95%
ETL Jobs	95%	98%
APIs	95%	98%

8. STEWARDSHIP WORKFLOW

8.1 Curation Process

Step 1: Scanner discovers new asset

Status: "DISCOVERED"

↓

Step 2: Steward receives alert

"New Assets Found"

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Step 3: Steward reviews technical metadata

Validates accuracy

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Step 4: Steward adds business metadata

- Link Business Terms

- Add descriptions

- Assign classifications

Status: "IN_REVIEW"

↓

Step 5: Data Owner reviews

Validates business context

↓

Step 6: Owner approves

Status: "CERTIFIED"

↓

Step 7: Asset published to catalog

Searchable by all users

8.2 Certification Levels

Level	Badge	Requirements
Bronze	Basic	Technical metadata complete
Silver	Standard	Business metadata complete
Gold	Advanced	Full metadata + lineage
Platinum	Premium	Gold + quality metrics + steward

8.3 SLA for Curation

Activity	SLA
Initial review	5 business days
Business enrichment	10 business days

Owner approval	5 business days
Total time to certification	20 business days

APPENDIX A: MANDATORY METADATA ATTRIBUTES

Tables

Attribute	Required	Example
Name	Yes	TBL_PATIENT_DEMOGRAPHICS
Description	Yes	"Contains patient demographic information"
Database	Yes	EPIC_PROD
Schema	Yes	CLINICAL
Data Owner	Yes	Dr. Sarah Al-Qahtani
Data Steward	Yes	Layla Al-Otaibi
Classification	Yes	CONFIDENTIAL
PII Indicator	Yes	YES
Source System	Yes	Epic EHR
Row Count	Yes	18,500,000

Columns

Attribute	Required	Example
Name	Yes	PATIENT_FIRST_NAME
Description	Yes	"Patient's legal first name"
Data Type	Yes	VARCHAR2(100)
Nullable	Yes	NO
Business Term Link	Recommended	Patient Name
PII Indicator	Yes	YES
Null Percentage	Yes	0.1%

APPENDIX B: METADATA QUALITY SCORECARD

Monthly Scorecard Template

Metric	Target	Actual	Status
Tables with descriptions	95%	___%	_____
Columns with descriptions	80%	___%	_____
Assets with owners	100%	___%	_____
Assets with stewards	100%	___%	_____
Assets with classification	100%	___%	_____
Lineage documented	90%	___%	_____
New assets curated < 20 days	95%	___%	_____

Document End

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